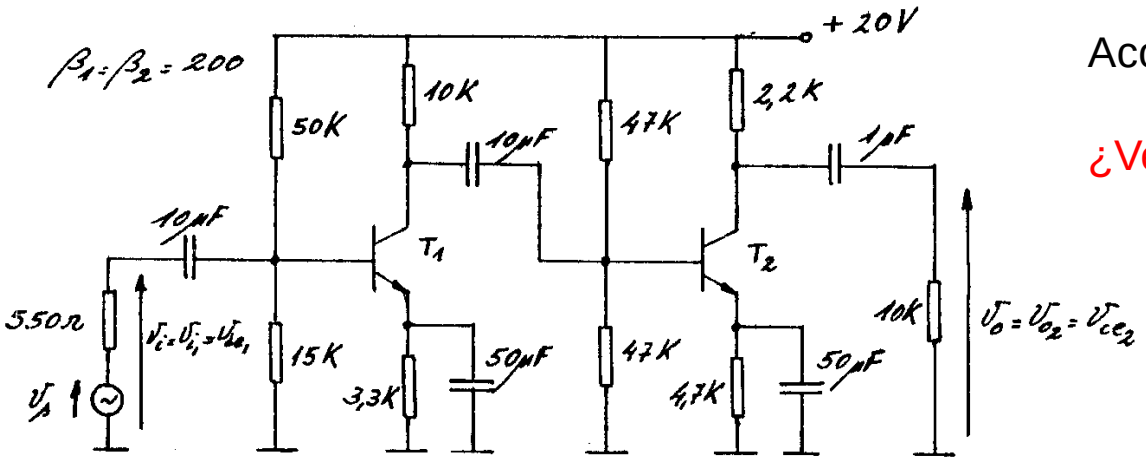
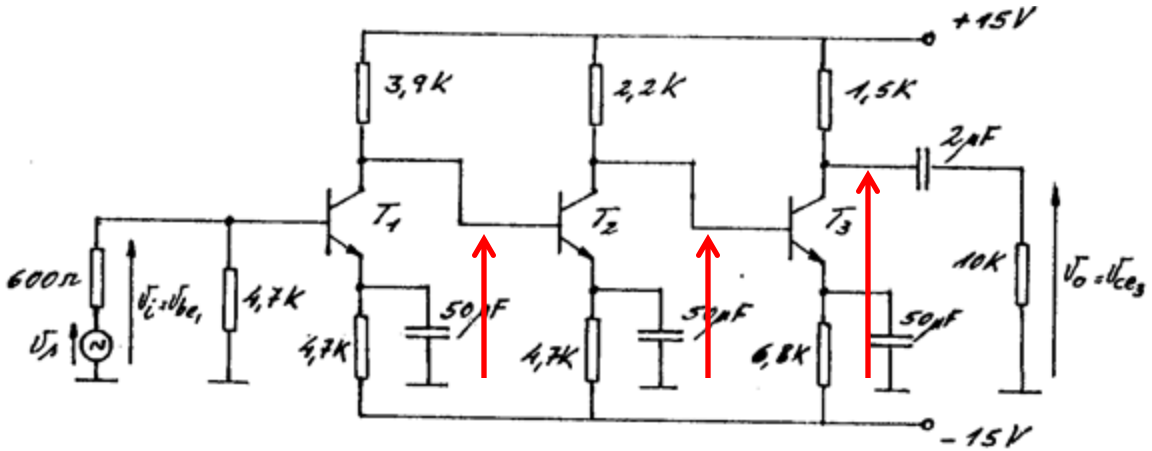
Amplificadores con varios transistores



Acople capacitivo entre etapas:

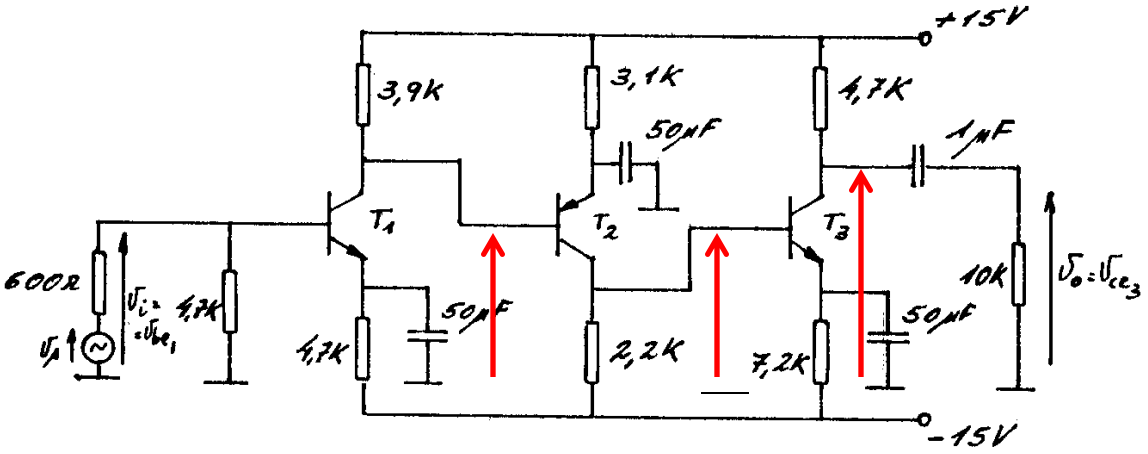
¿Ventajas? ¿Desventajas?



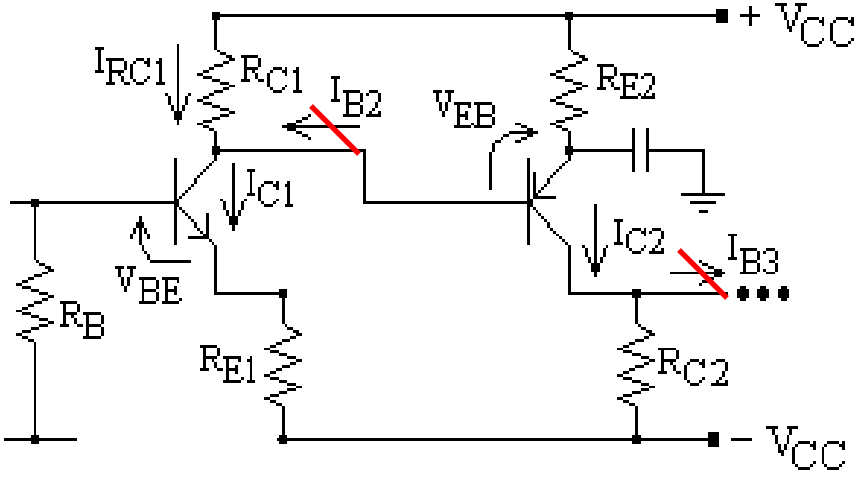
Acople directo entre etapas:

¿Qué pasa con V_{CQ}?

Solución:



¿Cómo se resuelve Q?:



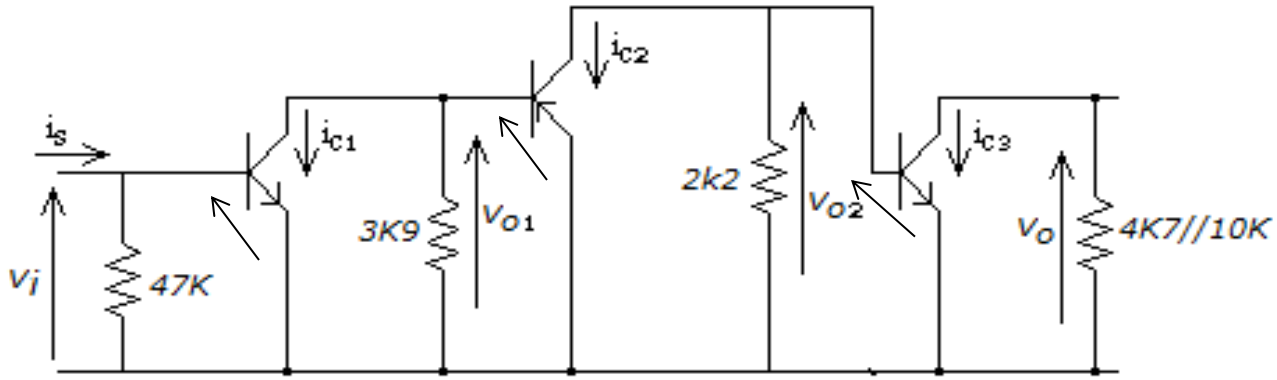
Estimación:

IC2máx = 2VCC /(RE2+RC2)

Si IC2máx /β << ICQ1

Entonces ~~*IBQ2*~~

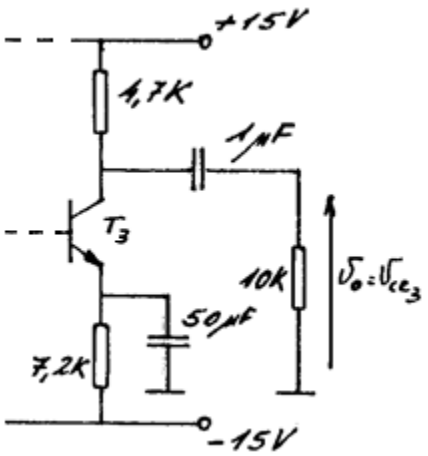
Análisis en señal (a frecuencias medias):

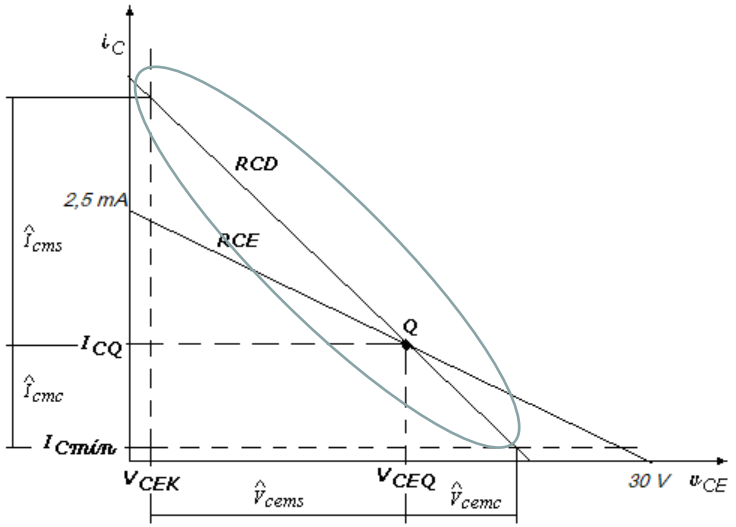


Etapa 1 **Etapa 2** **Etapa 3**

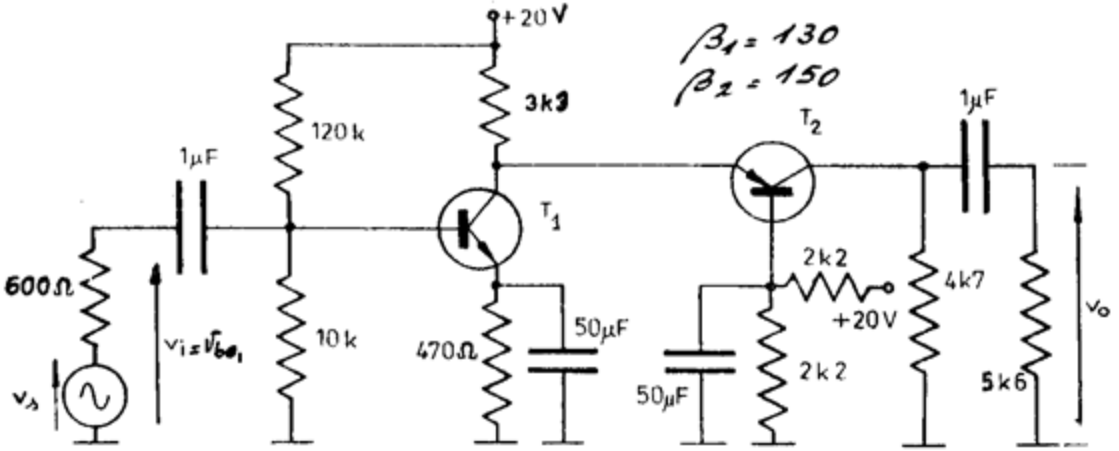
$$\frac{vo1}{vi} = -\frac{ic1(3K9 \,//\, r\pi 2)}{vbe1} \;\ldots\ldots\; \rightarrow \frac{vo}{vi} = \frac{vo1}{vi}\,\frac{vo2}{vo1}\,\frac{vo}{vo2} = Av$$

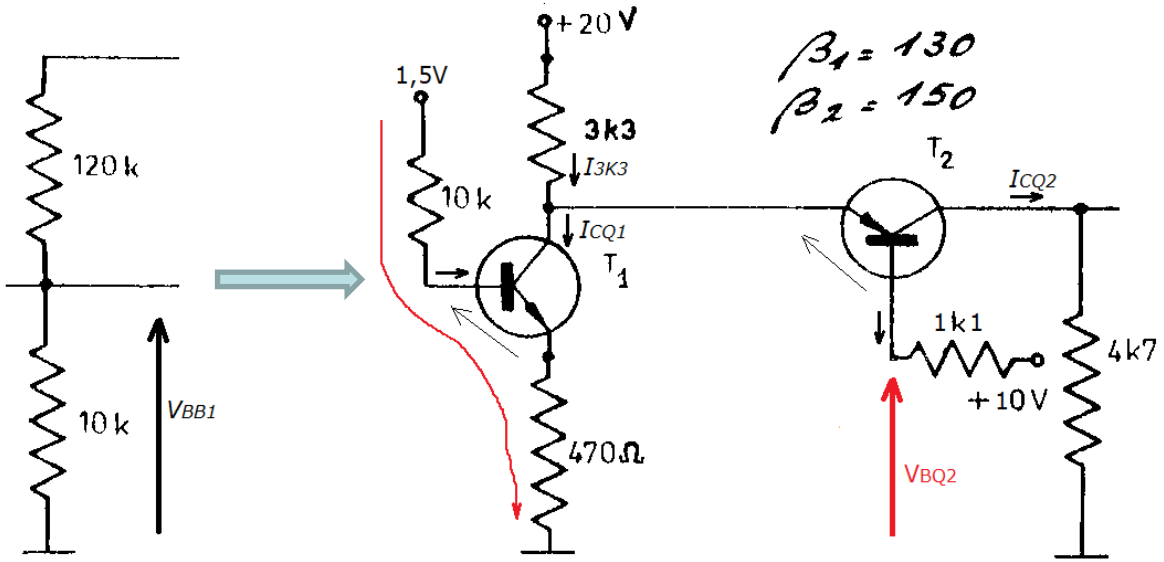
Una aclaración…

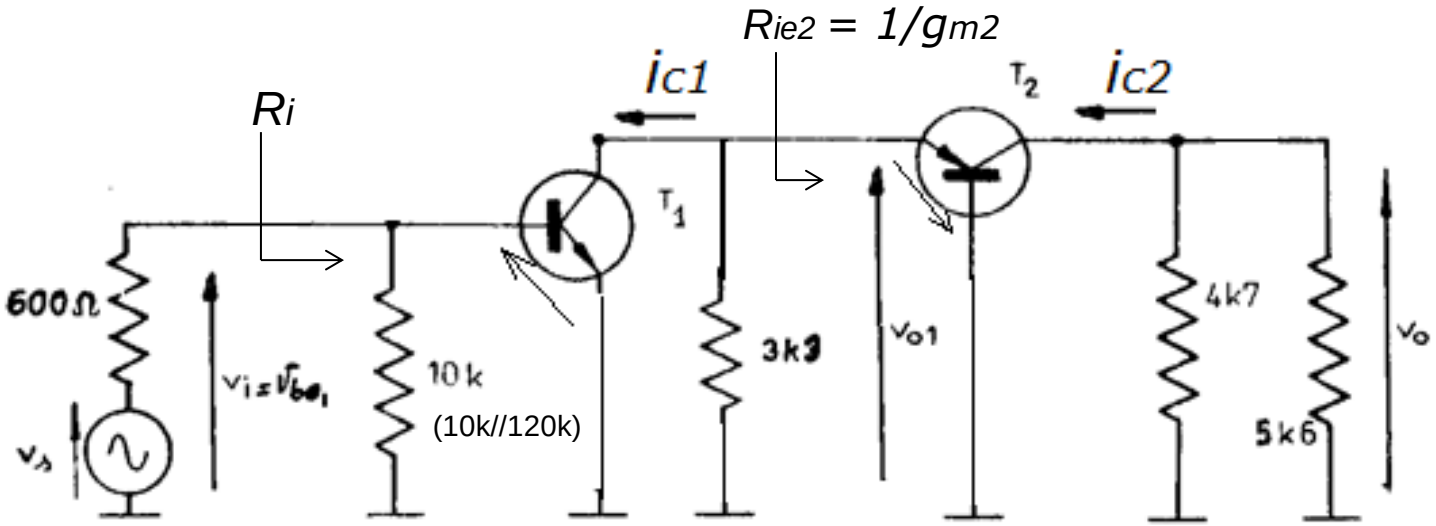




Otro ejemplo:







$I_{CQ1} \cong I_{CQ2} \cong 1{,}4\ mA \longrightarrow g_m \cong 56\ mA/V$

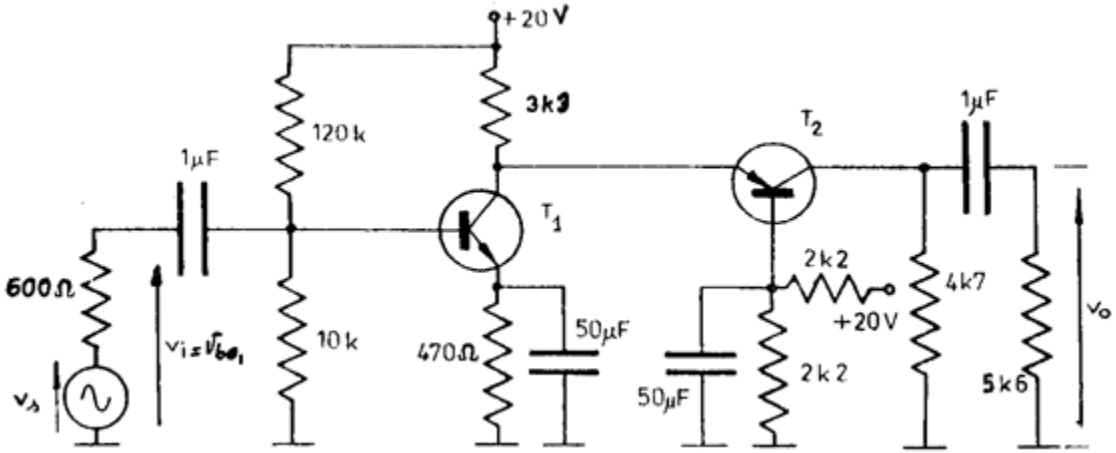
$V_{CEQ1} \cong 10\ V\ ;\ V_{ECQ2} \cong 4\ V \quad r_{\pi 1} \cong 2{,}3K\Omega\ ;\ r_{\pi 2} \cong 2{,}7K\Omega$

$$\frac{v_{o1}}{v_i} = -\frac{i_{c1}(3K3 \,//\, 1/g_{m2})}{v_{be1}} \cong -\, g_{m1}/g_{m2} \cong -1$$

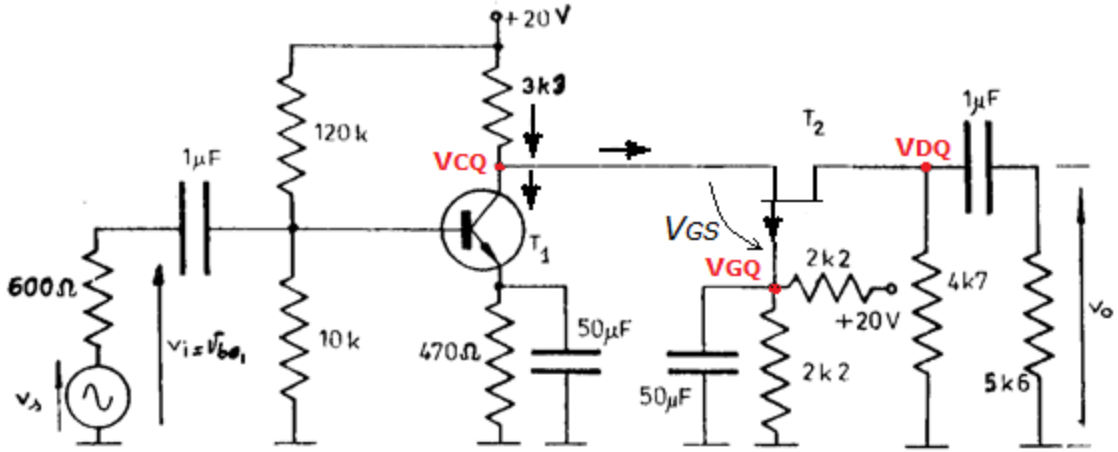
$$\frac{v_o}{v_{o1}} = -\frac{i_{c2}(4K7 \,//\, 5K6)}{-\, v_{be2}} \cong g_{m2}\,(4K7 \,//\, 5K6) \cong 140$$

$$\frac{v_o}{v_i} = g_{m1}\,(4K7 \,//\, 5K6) = A_v \cong -\,140$$

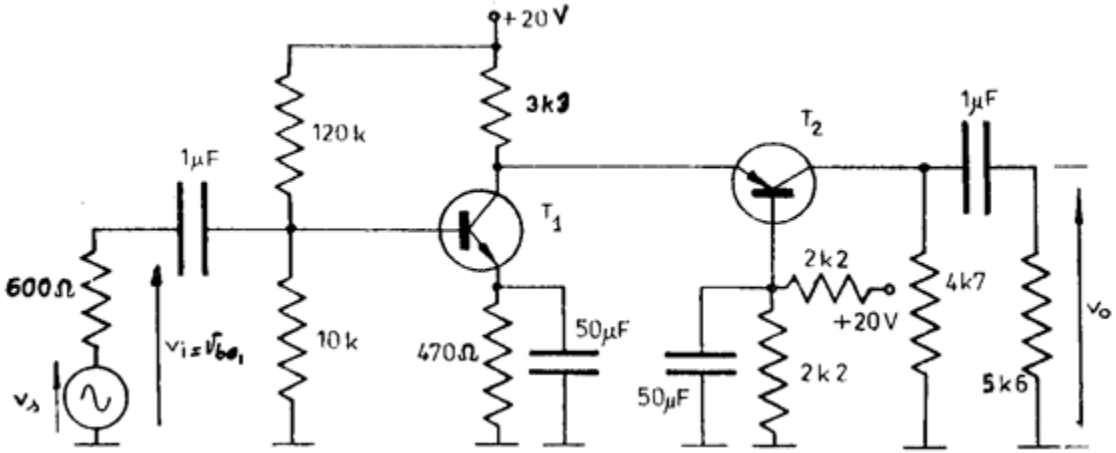
¿Qué pasa si…? (análisis cualitativo):



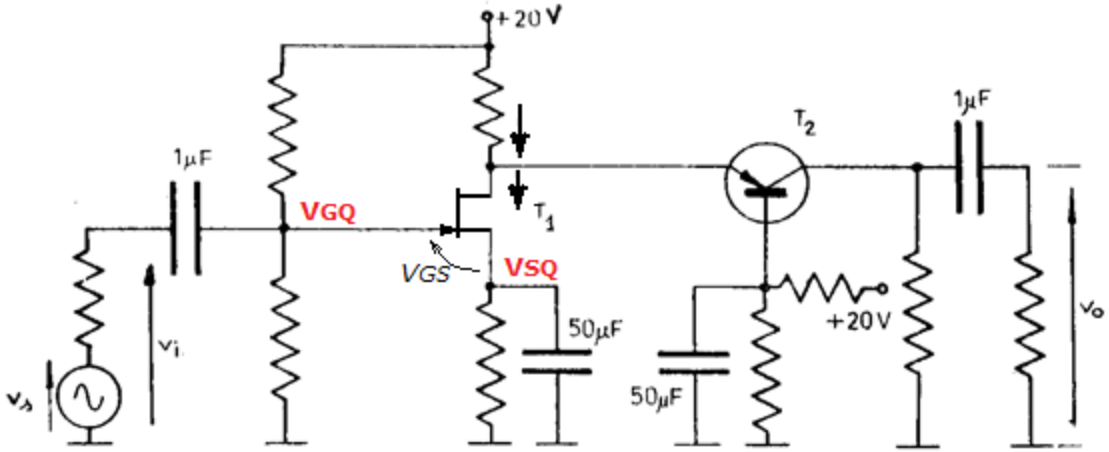
…se reemplaza T2 por un JFET…



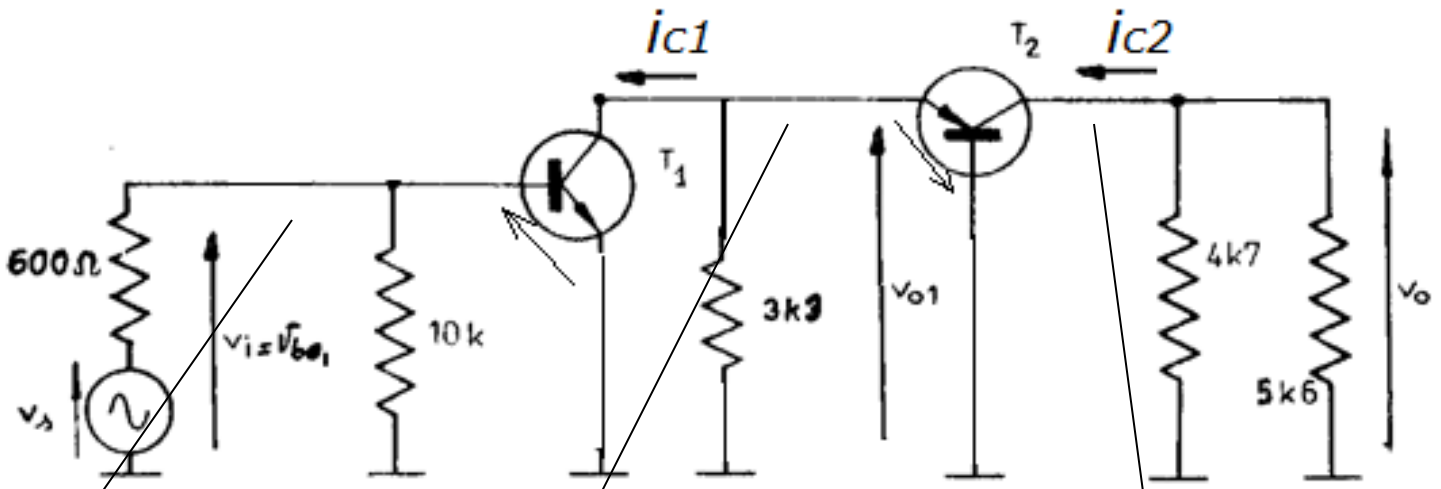
¿Qué pasa si…? (análisis cualitativo):



…se reemplaza T1 por un JFET…



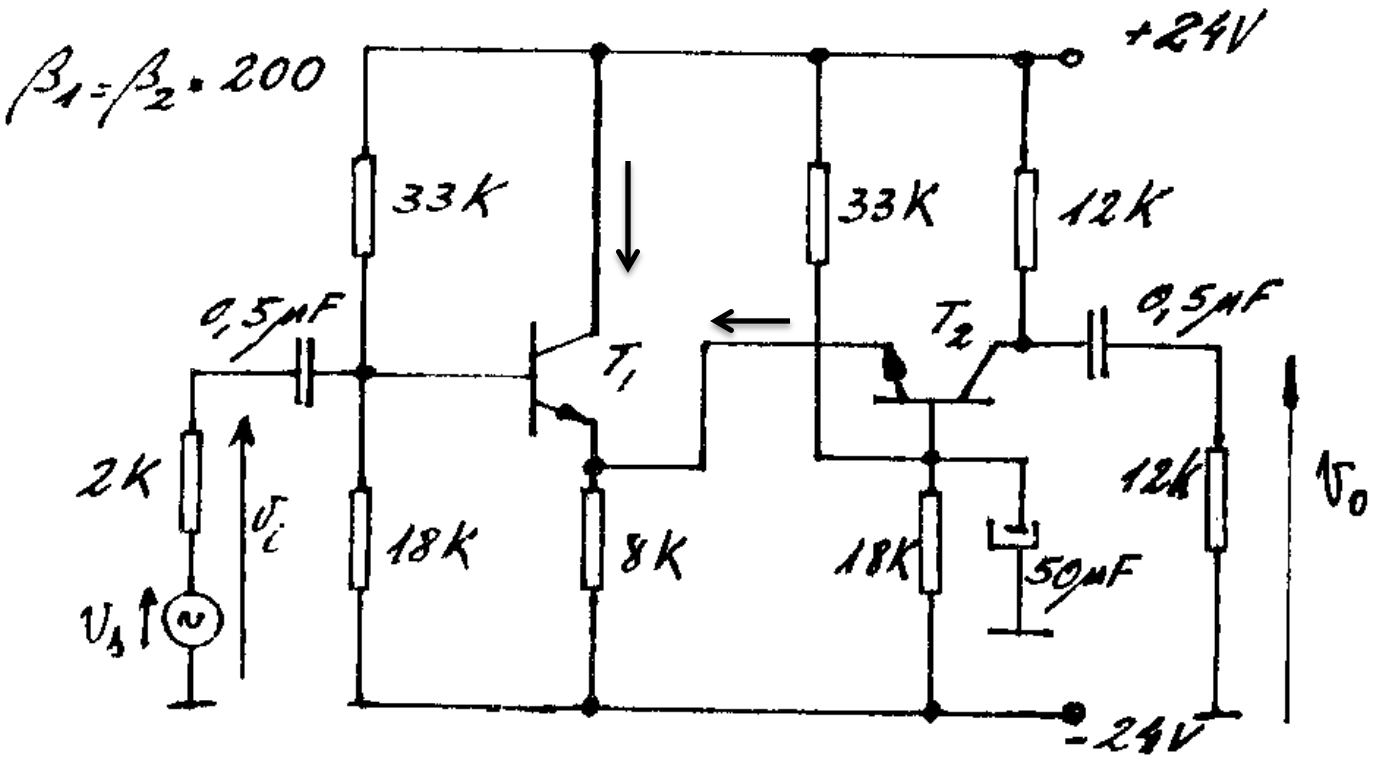
Un análisis básico de comportamiento al ruido…



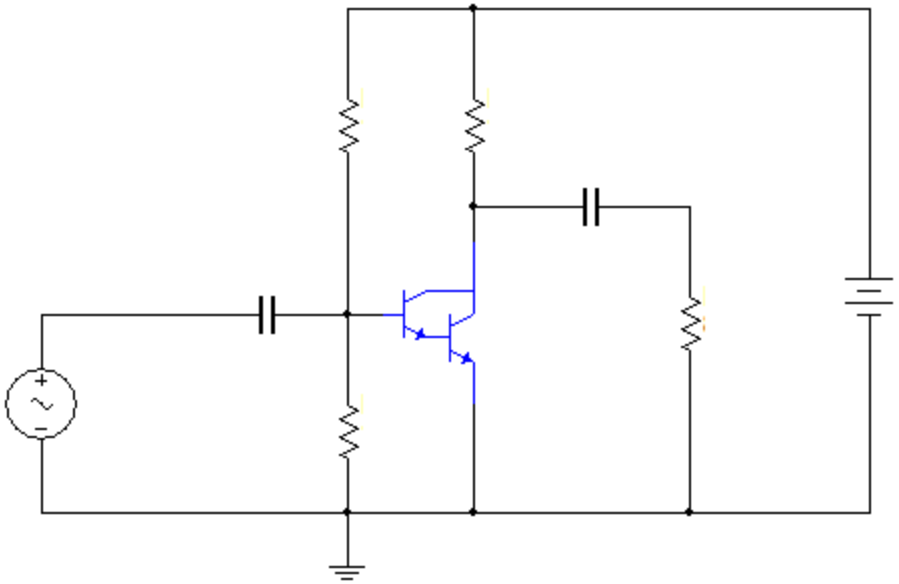
Vin + Vrin Vin Av1 + (Vrin Av1 + VrT1) Vin Av1 Av2 + (Vrin Av1 Av2 + VrT1 Av2 + VrT2)

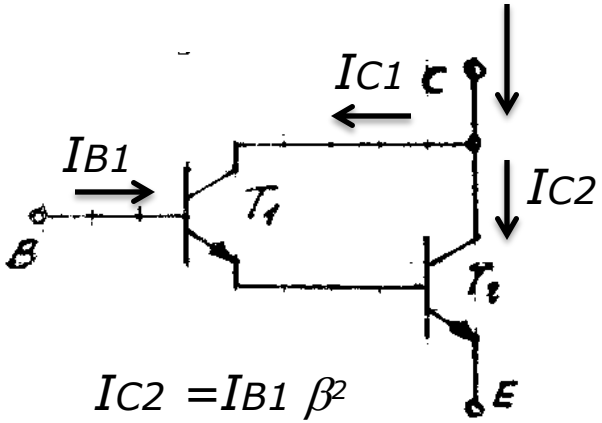
Si = v_{in}^2 / v_{rin}^2 > So = v_o^2 / v_{ro}^2

Otro ejemplo…



Otro ejemplo: Darlington





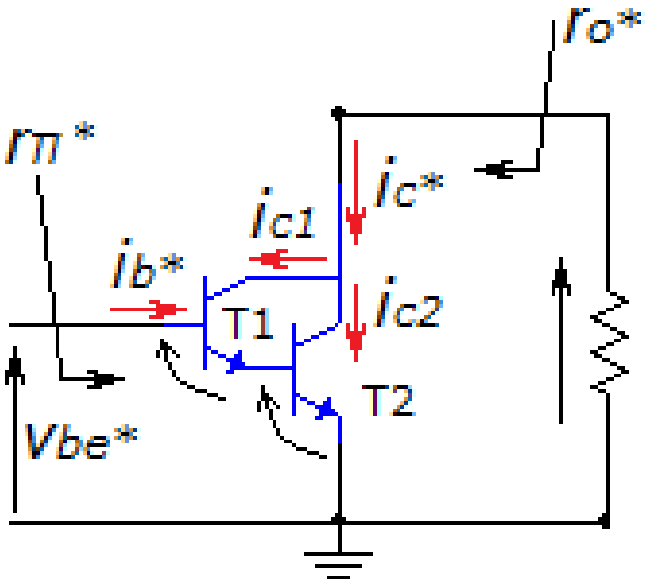
$I_{C2} = I_{B1}\,\beta^2$

Por TBJ...

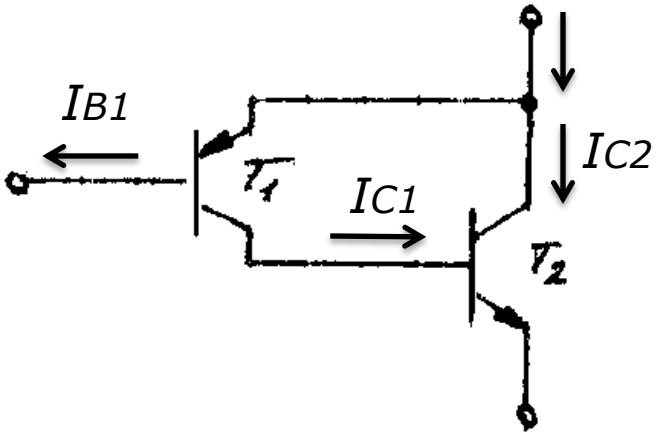
$r_\pi^* = r_{\pi 1} + \beta_1 r_{\pi 2} = \mathbf{2\, r_{\pi 1}}$

$g_{m^*} = i_c^*/v_{be}^* = i_{c2} / 2\, v_{be2} = \mathbf{g_{m2} / 2}$

$r_o^* = \mathbf{2\, r_{o2} / 3}$



Otros ejemplos de Darlington: cuasi-Darlington



IGBT:

